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import cv2
import numpy as np
import matplotlib.pyplot as plt
import time

# Read image
img = cv2.imread("images.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Different feature limits
feature_counts = [100, 300, 500, 800]

results = []

for n in feature_counts:

    orb = cv2.ORB_create(nfeatures=n)

    start = time.time()
    keypoints, descriptors = orb.detectAndCompute(gray, None)
    end = time.time()

    detected = len(keypoints)
    processing_time = (end - start) * 1000

    results.append([n, detected, processing_time])

    output = cv2.drawKeypoints(
        img, keypoints, None,
        color=(0,255,0),
        flags=cv2.DrawMatchesFlags_DRAW_RICH_KEYPOINTS
    )

    cv2.imwrite(f"ORB_{n}.jpg", output)

print("Requested\tDetected\tTime(ms)")
```

```
for r in results:  
    print(r[0], "\t\t", r[1], "\t\t", round(r[2],2))
```

Requested	Detected	Time(ms)
100	100	43.76
300	300	15.79
500	500	16.18
800	800	17.09